

TrueNAS SCALE on Proxmox

HBA passthrough from BIOS to a running pool

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2026-09-10

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Preface

This book is a working runbook for running **TrueNAS SCALE as a VM on a Proxmox VE host**, with the data disks attached through a dedicated **HBA in IT mode** passed through to the guest.

That is the layout iXsystems still recommends for production data: the hypervisor owns its own boot disk; TrueNAS owns the storage controller and the pool.

Use it as:

- a **Quarto book** (numbered chapters, cross-references, PDF/EPUB) — download the PDF
- a **website** (`quarto render → _book/index.html`, then publish with `quarto publish`)

How to build

In RStudio: **File → Open Project** on this folder, then **Render Book**.

From a terminal in this directory:

```
quarto preview
quarto render
quarto publish gh-pages
```

Scope

Chapters walk the host from a cold BIOS through IOMMU, vfio binding, the TrueNAS VM, the PCI passthrough line, and the failures that usually show up after the HBA is attached.

Commands assume a current Proxmox VE 8/9 host and TrueNAS SCALE 24.10 / 25.10. PCI addresses and vendor:device IDs in examples are placeholders. Replace them with output from your own `lspci`.

Chapter 1

Overview

1.1 Target architecture

```
[ motherboard BIOS: VT-x/SVM + VT-d/AMD-Vi ]
      ↓
[ Proxmox kernel: intel/amd_iommu=on iommu=pt + vfio ]
      ↓
[ HBA bound to vfio-pci, blacklisted from host SAS driver ]
      ↓
[ TrueNAS VM: q35 + OVMF + hostpci0 ...,pcie=1,rombar=0 ]
      ↓
[ TrueNAS owns the HBA and the data pool ]
```

Proxmox boots from its own NVMe or board SATA. TrueNAS boots from a small virtual disk. The HBA and every disk on it belong only to the VM.

1.2 Why pass the HBA, not individual disks

Virtual disks and “passthrough by serial” sit behind the hypervisor. ZFS then cannot trust SMART, cache flush, or device identity the way it can on bare metal. Passing the **entire controller** gives TrueNAS the same view it would have on metal: serials, SMART, resilver, and disk replace.

Do **not** put the Proxmox boot drive on that HBA.

1.3 Hardware checklist

- CPU and board with IOMMU: Intel **VT-d** or AMD **AMD-Vi**
- Dedicated HBA flashed to **IT mode** (not IR/RAID)

- Cards that behave well: LSI/Broadcom 9207-8i, 9300-8i, 9400-8i, Dell H310 in IT mode
- Confirm firmware with `sas2flash -list` or `sas3flash -list` — look for (IT)
- RAM for the VM: 16 GB is a floor; more if the pool is large
- HBA in a PCIe slot that can land in its own IOMMU group (often slot 1 on consumer boards)

 Warning

IR/RAID firmware hides disks behind a RAID abstraction. ZFS needs raw devices. Flash IT mode **before** you pass the card through.

Chapter 2

BIOS and UEFI settings

Reboot the physical host and enter setup (Del, F2, or F10). Names vary by vendor. Enable every row that exists.

2.1 Settings that matter

Setting	Set to	Why
Intel VT-x or AMD SVM / AMD-V	Enabled	Basic virtualization. Not IOMMU.
Intel VT-d or AMD AMD-Vi / IOMMU	Enabled	Required for PCI passthrough.
SR-IOV	Enabled if present	Not required for an HBA.
Above 4G Decoding	Enabled	More important for GPUs; enable anyway.
Resizable BAR	Off unless you also pass a GPU	Can confuse some HBAs.
CSM / Legacy boot	Off if Proxmox is UEFI	Stay in UEFI end to end.
Secure Boot	Off (simplest)	Avoids vfio module signing fights.
iGPU / Integrated Graphics	Disable if the VM hangs after adding the HBA	Common Intel hang fix.

Setting	Set to	Why
Slot Option ROM / OPROM for the HBA	Disabled if available	Stops the card from trying to boot the host.

Save and reboot into Proxmox.

2.2 Where vendors hide IOMMU

- **Intel:** Advanced → CPU Configuration, System Agent, or Chipset. Labels: *VT-d, Intel VT for Directed I/O, IOMMU*.
- **AMD:** Advanced → AMD CBS → NBIO, or next to *SVM Mode*. Labels: *IOMMU, AMD-Vi*.

! Important

VT-x / SVM alone is not enough. Without VT-d or AMD-Vi, `/sys/kernel/iommu_groups` stays empty and Proxmox will refuse the PCI device.

2.3 Slot choice

On many B450/B550/X570 boards the HBA only isolates cleanly in the **first full-length PCIe slot**. If Chapter 4 shows a shared group, move the card before you add ACS override.

Chapter 3

Enable IOMMU on the Proxmox host

SSH into the host. See how it boots:

```
proxmox-boot-tool status
```

3.1 GRUB

Typical on BIOS installs and some older UEFI installs. Edit `/etc/default/grub`.

Intel:

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet intel_iommu=on iommu=pt"
```

AMD (modern kernels often enable AMD-Vi without the extra flag; keeping it is fine):

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet amd_iommu=on iommu=pt"
```

Apply:

```
update-grub
```

3.2 systemd-boot

Typical on current UEFI Proxmox 8/9. Edit `/etc/kernel/cmdline` and **append** to the existing line. Do not delete `root=`.

Intel:

```
intel_iommu=on iommu=pt
```

AMD:

```
amd_iommu=on iommu=pt
```

Apply:

```
proxmox-boot-tool refresh
```

`iommu=pt` is passthrough mode: only devices given to VMs pay IOMMU cost.

i Note

On Proxmox kernels 6.8+, Intel often turns IOMMU on without `intel_iommu=on`. Keep the flag anyway so the intent is obvious in `/proc/cmdline`.

3.3 Load VFIO at boot

Add to `/etc/modules`:

```
vfio
vfio_iommu_type1
vfio_pci
```

`vfio_virqfd` was folded into other modules on newer kernels. Skip it if `modprobe` complains.

Reboot, then check:

```
dmesg | grep -e DMAR -e IOMMU -e AMD-Vi
cat /proc/cmdline
```

You want DMAR or AMD-Vi tables found, “IOMMU enabled”, and the flags in the command line.

Chapter 4

Identify the HBA and IOMMU group

4.1 Find the card

```
lspci -nn | grep -iE 'sas|scsi|lsi|broadcom|aha|hba'
```

Example output:

```
02:00.0 Serial Attached SCSI controller [0107]: Broadcom / LSI SAS3008 [1000:0097]
```

- 02:00.0 is the PCI address
- 1000:0097 is vendor:device (needed for vfio)

4.2 Print IOMMU groups

```
for d in /sys/kernel/iommu_groups/*/devices/*; do
  g=${d#/iommu_groups/}; g=${g%/*}
  echo "IOMMU Group $g: $(lspci -nns ${d##*/})"
done | sort -V
```

The HBA should sit **alone** in its group, or only with its own extra functions (02:00.0, 02:00.1, ...).

4.3 When the group is dirty

If the HBA shares a group with the management NIC, GPU, chipset USB, or the Proxmox boot controller, do not pass that group. The host will lose those devices the moment the VM starts.

Fix in this order:

1. Move the HBA to another PCIe slot (try slot 1).
2. Update the motherboard BIOS.
3. Last resort on a single-tenant homelab only: add to the kernel command line:

```
pcie_acs_override=downstream,multifunction
```

Then `update-grub` or `proxmox-boot-tool refresh` and reboot.

 Warning

ACS override weakens DMA isolation between devices. Fine on a box only you use. Do not use it on a shared host.

Chapter 5

Bind the HBA to vfio-pci

If Proxmox loads `mpt3sas`, `mpt2sas`, or `megaraid_sas` on the card, the host can see — and touch — the TrueNAS disks. Bind the card to `vfio` before you create the VM.

5.1 Confirm IDs

```
lspci -nn -s 02:00.0
```

Use **your** address and ID. The rest of this chapter keeps `02:00.0` and `1000:0097` as the example.

5.2 Bind and blacklist

```
echo "options vfio-pci ids=1000:0097" > /etc/modprobe.d/vfio.conf
echo "blacklist mpt3sas" >> /etc/modprobe.d/blacklist.conf
echo "blacklist mpt2sas" >> /etc/modprobe.d/blacklist.conf
update-initramfs -u -k all
reboot
```

After reboot:

```
lspci -k -s 02:00.0
```

Kernel driver in use: must be `vfio-pci`, not `mpt3sas`. The data disks should vanish from the Proxmox **Disks** view.

5.3 If the host driver still wins

```
echo "softdep mpt3sas pre: vfio-pci" >> /etc/modprobe.d/vfio.conf
update-initramfs -u -k all
reboot
```

Multi-function cards: put every vendor:device ID on one line, comma-separated:

```
options vfio-pci ids=1000:0097,1000:00c9
```


Chapter 6

Create the TrueNAS SCALE VM

Install TrueNAS onto a **virtual OS disk** first, with no HBA attached. Confirm it boots. Add passthrough in Chapter 7.

6.1 Settings that matter

Setting	Value
Machine	q35
BIOS	OVMF (UEFI)
CPU type	host
Cores	4 or more
RAM	16 GB minimum, ballooning off
SCSI controller (OS disk only)	VirtIO SCSI
OS disk	16–32 GB on Proxmox SSD/NVMe
Network	VirtIO on vmbr0
Start at boot	Off until passthrough works

q35 is required for real PCIe. i440fx only does legacy PCI.

6.2 Example `qm create`

```
qm create 300 \  
  --name truenas \  
  --memory 16384 \
```

```
--balloon 0 \  
--cores 4 \  
--cpu host \  
--bios ovmf \  
--machine q35 \  
--efidisk0 local-lvm:1,efitype=4m,pre-enrolled-keys=0 \  
--scsihw virtio-scsi-single \  
--scsi0 local-lvm:32,ssd=1 \  
--net0 virtio,bridge=vmbr0 \  
--ostype l26
```

Attach the SCALE ISO, set boot order to the CD, install onto `scsi0`, then set boot back to the virtual disk.

 Tip

Keep the TrueNAS boot pool on the virtual disk. The HBA is only for the data pool. You can rebuild the VM and import the same pool later.

Chapter 7

Pass the HBA into the VM

7.1 From the Proxmox UI

VM → **Hardware** → **Add** → **PCI Device**.

- Device: 02:00.0, or the whole 02:00 if you tick **All Functions**
- **PCI-Express**: on (pcie=1)
- **ROM-Bar**: off (rombar=0)
- Primary GPU: off

ROM-Bar off is the usual fix for a VM that hangs on “Booting from Hard Disk” after an LSI card is added.

7.2 From the CLI

One function:

```
qm set 300 --hostpci0 02:00.0,pcie=1,rombar=0
```

Every function at that address (preferred for HBAs):

```
qm set 300 --hostpci0 02:00,pcie=1,rombar=0
```

Do **not** make the HBA a boot device. TrueNAS boots from the virtual OS disk.

7.3 After start

In TrueNAS: **Storage** → **Disks**. You should see real model names, serials, and SMART. Import an existing pool or create a new one.

The matching line in `/etc/pve/qemu-server/300.conf` looks like:

```
hostpci0: 0000:02:00,pcie=1,rombar=0
```

Chapter 8

TrueNAS after passthrough

8.1 First checks

- Disks appear with serial numbers, not generic **sdX** only
- SMART works inside TrueNAS, not on the Proxmox host
- Pool import or create succeeds without “disk in use” errors

8.2 Guest hygiene

- Static MAC on the VirtIO NIC, plus a DHCP reservation or static IP
- Do not use Proxmox snapshots of the virtual OS disk as the backup plan for the pool
- Snapshot and replicate **inside** TrueNAS (or **zfs send** to another box)
- VirtIO networking is enough for SMB/NFS. Pass a spare NIC later if you want 10 GbE with less CPU

8.3 What not to do

- Do not let Proxmox mount or SMART-scan the HBA disks
- Do not put swap or the Proxmox boot disk on the passed-through card
- Do not enable QEMU disk cache tricks on the data pool — there is no QEMU disk in front of it

Chapter 9

Troubleshooting

9.1 VM hangs at “Booting from Hard Disk”

Set `rombar=0` (uncheck **ROM-Bar**). Disable the slot Option ROM in system BIOS. Some LSI utilities also have “boot from this controller” — turn that off. The VM should boot the virtual OS disk only.

9.2 IOMMU groups missing / “PCI device not available”

VT-d or AMD-Vi is still off, or the kernel flags did not apply.

```
cat /proc/cmdline  
dmesg | grep -e DMAR -e IOMMU -e AMD-Vi
```

Revisit Chapter 2 and Chapter 3.

9.3 Proxmox still lists the data disks

`vfio` bind failed. `lspci -k -s 02:00.0` should show `vfio-pci`. Rebuild `initramfs` after blacklist changes (Chapter 5).

9.4 TrueNAS sees the HBA but no disks

- Confirm IT firmware, not IR
- Pass **all functions** of the card
- Reseat SFF-8087 / 8643 cables; the card can enumerate with no SAS link
- Check disk power

9.5 Host dies when the VM autostarts

The HBA shared an IOMMU group with the management NIC. From the boot menu, drop `intel_iommu=on` / `amd_iommu=on` for one boot, turn VM autostart off, fix grouping (Chapter 4), then re-enable IOMMU.

9.6 Need ACS override?

Only after a slot change and a BIOS update still leave a dirty group. Homelab only.

9.7 Quick verification set

```
cat /proc/cmdline
dmesg | grep -e DMAR -e IOMMU -e vfio
lspci -k -s 02:00.0
ls /sys/kernel/iommu_groups
```


Chapter 10

Command reference

10.1 Host

```
proxmox-boot-tool status
cat /proc/cmdline
dmesg | grep -e DMAR -e IOMMU -e AMD-Vi -e vfio
lspci -nn | grep -iE 'sas|scsi|lsi|broadcom'
lspci -k -s 02:00.0
```

IOMMU groups:

```
for d in /sys/kernel/iommu_groups/*/devices/*; do
  g=${d#/iommu_groups/}; g=${g%/*}
  echo "IOMMU Group $g: $(lspci -nns ${d##*/})"
done | sort -V
```

10.2 Kernel flags

Intel GRUB:

```
quiet intel_iommu=on iommu=pt
```

AMD GRUB:

```
quiet amd_iommu=on iommu=pt
```

Optional last-resort ACS split:

```
pcie_acs_override=downstream,multifunction
```

10.3 vfio

```
# /etc/modprobe.d/vfio.conf
options vfio-pci ids=1000:0097
softdep mpt3sas pre: vfio-pci

# /etc/modprobe.d/blacklist.conf
blacklist mpt3sas
blacklist mpt2sas

update-initramfs -u -k all
```

10.4 VM

```
qm set 300 --hostpci0 02:00,pcie=1,rombar=0
```

/etc/pve/qemu-server/300.conf fragment:

```
bios: ovmf
machine: q35
cpu: host
balloon: 0
hostpci0: 0000:02:00,pcie=1,rombar=0
```

Replace 02:00 and 1000:0097 with your `lspci -nn` output.